

ENGR 213: Applied Ordinary Differential Equations

Paradis notes — In-Class Lecture, Tutorial & Midterm 1 Preparation Master Compendium

Concordia University · Gina Cody School of Engineering · Department of MIAE

Curriculum Authority: Dr. Alexandre Paradis, ing. Ph.D. · In-Class Lecture Notes 1–5 · Tutorials 1 & 3 · Homework Sets 1–3

Examination Target: Midterm #1 (October 19, 2026) · **Scope:** Chapter 2 (Sections 2.1–2.8) + Chapter 17 (Sections 17.1 & 17.2)

ENGR 213: Applied Ordinary Differential Equations

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Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

****Academic Session:**** Fall 2026 / Winter 2025 · ****Course:**** ENGR 213

****Instructor:**** Dr. Alexandre Paradis, ing. Ph.D.

****Curriculum Mapping:**** Lectures 1–5 · Tutorials 1 & 3 · Homework Sets 1–3 · Midterm #1 Examination Scope (Oct 19: Chapter 2 + §17.1 & §17.2)

Executive Blueprint & Pedagogical Architecture

This document provides a regenerated, rigorously structured, and comprehensive compendium of Dr. Alexandre Paradis's in-class lectures, weekly tutorials, assigned textbook homework sets, and Midterm #1 examination preparation. All handwritten whiteboard and OneNote derivations have been converted into standard mathematical typesetting with complete proofs, physical interpretations, verification steps, and high-yield examination strategies.

ENGR 213 MIDTERM 1 CURRICULUM ARCHITECTURE (OCTOBER 19)

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[CHAPTER 1 & 2: FIRST-ORDER ODEs] [CHAPTER 17: COMPLEX VARIABLES]

- * §1.1: Order, Linearity & Intervals of Definition * §17.1: Complex Arithmetic & Cartesian Form
- * §2.1: Direction Fields & Autonomous Critical Points * §17.2: Polar & Exponential Form (Euler's)
- * §2.2: Separable Equations & Initial Value Problems * De Moivre's Theorem & n-th Roots
- * §2.3: First-Order Linear Equations (Integrating Factor) * Characteristic Roots of 2nd-Order ODEs
- * §2.4: Exact Differential Equations & Integrating Factors
- * §2.5: Solutions by Substitution (Bernoulli, Homogeneous, Linear)
- * §2.7: Linear Models (Radioactive Decay, Newton's Cooling)
- * §2.8: Nonlinear Models (Logistic Growth, Torricelli's Law)

Module 1: Fundamental Concepts & Classification (Lecture 1)

****Date Delivered:**** September 8, 2026 · ****Textbook Reference:**** Zill, Sections 1.1, 1.2 & 2.1

1.1 Definition of Ordinary Differential Equations (ODEs)

A differential equation is a mathematical relation containing one or more dependent variables (functions) and one or more of their derivatives with respect to one or more independent variables.

- **Ordinary Differential Equation (ODE):** Involves derivatives with respect to a single independent variable (e.g., x or t):

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- **Partial Differential Equation (PDE):** Involves partial derivatives of a multivariable function with respect to two or more independent variables:

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1.2 Standard Derivative Notations

- **Leibniz Notation:** Math input error (essential for separation of variables and integrating factors).
- **Prime Notation:** y' , y'' , y''' , $y^{(4)}$, $\dots$, $y^{(n)}$ (compact for linear constant-coefficient equations).
- **Newton Dot Notation (Time Derivatives):** Math input error (predominantly used in dynamics, vibrations, and mechatronics).

1.3 Order of a Differential Equation

The **order** of an ODE is defined strictly as the order of the highest derivative present in the equation:

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1.4 Linearity Criteria

An n -th order ODE is **linear** if it can be written in the form:

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The Two Golden Rules of Linearity:

1. The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of the **first power only** (no exponents like $(y')^2$ or y^3).
2. The coefficients $a_n(x), \dots, a_0(x)$ and the driving term $g(x)$ depend **solely on the independent variable x** (no products such as yy' or nonlinear functions like $\sin(y)$ or e^y).

Module 2: In-Class Tutorial 1 & Homework 1 Solutions

Date Delivered: September 14, 2026 · **Focus:** Solution Verification & Autonomous Phase Portraits

2.1 Problem §1.1 Q13: First-Order Linear Verification

- **Given ODE & Trial Solution:**

$$2y' + y = 0, \quad y(x) = e^{-x/2}$$

- **Step-by-Step Derivation:**

1. Differentiate the trial function: **Math input error**.
2. Substitute into the left-hand differential operator:

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3. The equation is identically satisfied for all $x \in \mathbb{R}$.

- **Suitable Interval of Definition:** $I = (-\infty, +\infty)$.

2.2 Problem §1.1 Q17: Radical Nonlinear Solution Verification

- **Given ODE & Trial Solution:**

$$(y - x)y' = y - x + 8, \quad y(x) = x + 4\sqrt{x+2}$$

- **Step-by-Step Derivation:**

1. Compute the derivative:

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2. Evaluate the Left-Hand Side (LHS):

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3. Evaluate the Right-Hand Side (RHS):

$$y - x + 8 = (x + 4\sqrt{x+2}) - x + 8 = 4\sqrt{x+2} + 8$$

4. LHS $\equiv$ RHS identically.

- **Domain & Interval of Validity:** For real numbers, the radicand requires $x + 2 \geq 0 \implies x \geq -2$. Since the derivative requires $\sqrt{x+2}$

eq 0\$ in the denominator, the solution exists on the open interval:

$$I = (-2, +\infty)$$

2.3 Problem §1.1 Q25: Second-Order Homogeneous Verification

- **Given ODE & Trial Solution:**

$$y'' - 4y' + 4y = 0, \quad y(x) = c_1e^{2x} + c_2xe^{2x}$$

- **Step-by-Step Derivation:**

1. Factor the trial function: $y = (c_1 + c_2x)e^{2x}$.

2. First derivative: $y' = c_2e^{2x} + 2(c_1 + c_2x)e^{2x} = (2c_1 + c_2 + 2c_2x)e^{2x}$.

3. Second derivative:

$$y'' = 2c_2e^{2x} + 2(2c_1 + c_2 + 2c_2x)e^{2x} = (4c_1 + 4c_2 + 4c_2x)e^{2x}$$

4. Substitute into the ODE:

Extra \left or missing \right

Extra \left or missing \right

- **Interval of Definition:** $I = (-\infty, +\infty)$.

2.4 Problem §2.1 Q21: Autonomous Direction Fields & Stability Analysis

- Given Autonomous ODE:

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- Phase Line & Critical Points:

Critical points occur where Math input error and $y_2 = 3$.

- Interval Sign Analysis:

* For $y < 0$: Math input error (solutions increase towards 0).

* For $0 < y < 3$: Math input error (solutions decrease towards 0).

* For $y > 3$: Math input error (solutions increase away from 3).

- Stability Classification:

* $y = 0$ is an Asymptotically Stable Attractor (Sink).**

* $y = 3$ is an Unstable Repeller (Source).**

Module 3: First-Order Solution Techniques (Lectures 2 & 3)

Dates Delivered: September 10 & 22, 2026 · **Textbook Reference:** Zill, Sections 2.2, 2.3 & 2.5

3.1 Separable Differential Equations (§2.2)

A first-order equation is separable if it can be written as:

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Standard 4-Step Solution Algorithm:

1. Separate variables: Math input error (provided $f(y)$

eq 0\$).

2. Integrate both sides: Math input error.

3. Check for singular solutions: Set $f(y) = 0$ to find equilibrium constant solutions that might not be contained in the family for any finite value of C .

4. Apply initial condition $y(x_0) = y_0$ to isolate C and solve explicitly for $y(x)$.

3.2 Dr. Paradis In-Class Quiz 1 Master Problem

- **Problem Statement:** Solve $yy' + x^2 = 4$, $y(0) = 2$.
- **Full Derivation:**

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Apply $y(0) = 2$:

$$2^2 = 8(0) - 0 + C_1 \implies C_1 = 4$$

Because $y(0) = +2$, choose the positive branch:

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3.3 First-Order Linear Equations & The Integrating Factor Method (§2.3)

Standard form:

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****Integrating Factor Theorem:****

Multiplying by $I(x) = e^{\int P(x) dx}$ converts the left-hand side into the exact derivative of a product:

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Integrating both sides:

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3.4 Solutions by Substitution (§2.5)

Dr. Paradis highlights three primary substitution techniques:

1. **Bernoulli Equations:**

$$y' + P(x)y = Q(x)y^n \quad (n$$

$$n \neq 0, 1)$$

$$\text{Substitution: Let } u = y^{1-n} \implies u' = (1-n)y^{-n}y'.$$

Dividing the ODE by y^n produces a linear ODE in u :

$$u' + (1-n)P(x)u = Q(x)$$

2. **Homogeneous Differential Equations of Degree Zero:**

$$M(x,y)dx + N(x,y)dy = 0$$

$$\text{Substitution: Let } u = \frac{y}{x}.$$

Substituted ODE separates immediately:

$$M(u)du + N(u)dv = 0$$

3. **Linear Substitutions:**

$$\frac{dy}{dx} = f(Ax + By + C) \quad (B$$

$$B \neq 0)$$

$$\text{Substitution: Let } u = Ax + By + C.$$

Separable form:

$$M(u)du = f(u)dx$$

Module 4: Exact Equations & Mathematical Modeling (Lectures 4 & Tutorial 3)

Dates Delivered: September 28 & 29, 2026 · **Textbook Reference:** Zill, Sections 2.4, 2.7 & 2.8

4.1 Exact Differential Equations (§2.4)

A differential expression $M(x, y)dx + N(x, y)dy = 0$ is an exact differential in a simply connected region R if:

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****Construction of Potential Function $\Psi(x, y)$:****

1. Integrate M with respect to x : $\Psi(x, y) = \int M(x, y) dx + g(y)$.
2. Differentiate with respect to y : **Math input error**.
3. Integrate $g'(y)$ to find $g(y)$ and set $\Psi(x, y) = C$.

4.2 Integrating Factors for Non-Exact Equations

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$

eq $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, *search for an integrating factor μ* :

* If **Math input error** (function of x only):

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* If **Math input error** (function of y only):

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4.3 Tutorial 3 Worked Problem 1: Radioactive Decay (§2.7)

- **Problem:** $A(0) = 100 \text{ g}$. After 6 hours, mass decreases by 3% (97% remains). Find $A(24)$.
- **Solution:**

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4.4 Tutorial 3 Worked Problem 2: Forensic Newton's Law of Cooling (§2.7)

- **Problem:** Normal temperature $T(0) = 98.6^{\circ}\text{ext}F$. Body discovered at time t_d at $85^{\circ}\text{ext}F$. One hour later ($t_d + 1$) at $80^{\circ}\text{ext}F$. Surrounding room temperature $T_s = 70^{\circ}\text{ext}F$.
- **Solution:**

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$$T(0) = 70 + C = 98.6 \implies C = 28.6 \implies T(t) = 70 + 28.6e^{kt}$$

At discovery t_d :

$$85 = 70 + 28.6e^{kt_d} \implies 28.6e^{kt_d} = 15$$

One hour later $t_d + 1$:

$$80 = 70 + 28.6e^{k(t_d+1)} \implies 28.6e^{kt_d}e^k = 10$$

Dividing equations:

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Determine t_d :

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4.5 Nonlinear Models: Logistic Growth & Torricelli's Law (§2.8)

- **Logistic Growth:**

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Carrying capacity is $K = a/b$. If $P < a/b$, population increases ($P' > 0$). If $P > a/b$, population decreases ($P' < 0$).

- **Torricelli's Law (Leaking Tank):**

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For cylindrical tank ($V = A_w h$):

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Module 5: Complex Numbers & Polar Representations (Lectures 4 & 5)

****Dates Delivered:**** September 29 & October 1, 2026 · ****Textbook Reference:**** Zill, Chapter 17 (Sections 17.1 & 17.2)

5.1 Cartesian Representation (§17.1)

A complex number $z \in \mathbb{C}$ is defined as:

$$z = x + iy \quad (i = \sqrt{-1}, i^2 = -1)$$

* Real Part: $\text{extRe}(z) = x$

* Imaginary Part: $\text{extIm}(z) = y$

* Complex Conjugate: $\overline{z}$

* Fundamental Identity: $z \overline{z} = |z|^2$

5.2 Polar Form & Geometric Modulus (§17.2)

Represented in the complex plane (Argand diagram):

- **Modulus (Absolute Value):**

$$|z| = r = \sqrt{x^2 + y^2}$$

- **Argument (Phase Angle in Radians):**

$$\theta = \arg(z)$$

****Crucial Rule:**** In ENGR 213, angles must always be in ****radians****, never degrees!

* Principal Argument $\text{extArg}(z) = \Theta \in (-\pi, \pi]$:

$$z = r e^{i\Theta}$$

- **Polar & Euler Representation:**

$$z = r(\cos \theta + i \sin \theta) = r e^{i\theta}$$

5.3 Complex Arithmetic in Polar Coordinates

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$.

1. **Multiplication:**

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Moduli multiply, arguments add.

2. **Division:**

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Moduli divide, arguments subtract.

3. **De Moivre's Theorem:**

$$z^n = (r e^{i\theta})^n = r^n e^{in\theta} = r^n [\cos(n\theta) + i \sin(n\theta)]$$

4. **n -th Roots of Complex Numbers:**

The n distinct roots of $w = r e^{i\theta}$ are given by:

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Midterm #1 Examination Strategic Checklist (October 19)

- Separation of Variables:** Isolate dy and dx , integrate, and resolve arbitrary constants immediately with initial conditions.
- Integrating Factor Checklist:** Always ensure the coefficient of y is $+1$ before calculating $P(x)$ and $I(x) = e^{\int P dx}$.
- Exactness Verification:** Always write $M(x,y)$ and $N(x,y)$ clearly before integrating.
- Bernoulli Substitution:** Memorize $u = y^{1-n}$ and divide by y^n first.
- Complex Numbers:** Keep angles in radians; use conjugates to divide Cartesian fractions.

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